

DATA ANALYTICS · PROJECT REPORT

# VENDOR PERFORMANCE ANALYSIS

*An end-to-end analysis of vendor sales, profitability  
and inventory performance*

SQL · PYTHON (EDA) · POWER BI

REPORT SNAPSHOT

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| Scope          | Vendor, purchase, sales, and inventory performance across the consolidated vendor_sales_summary dataset  |
| Analysis Tools | SQL (data preparation) · Python – pandas / seaborn / statistical testing (EDA) · Power BI (dashboarding) |
| Coverage       | 116 vendors · ~6,000 brands · four underlying source tables merged into one analytical table             |

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## SECTION 01

# Business Problem

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*The challenge driving this analysis*

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

## SECTION 02

# Data Overview

*Where the numbers in this report come from*

The analysis draws on a single aggregated table, **vendor\_sales\_summary**, which combines every vendor and business-related dataset used in this study. The four underlying sources are summarised below.

| DATASET                | DESCRIPTION                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>purchase_prices</b> | Purchase prices of different products and brands across vendors.                                                             |
| <b>purchases</b>       | Products purchased from each vendor, including quantities, prices, and transaction details.                                  |
| <b>vendor_invoice</b>  | Vendor invoices — purchase orders, invoice dates, quantities, purchase amounts, freight costs, and payment details.          |
| <b>sales</b>           | Sales of vendor-supplied products, including quantities, amounts, selling prices, dates, and related product/vendor details. |

All four sources are stored in a database for easy retrieval, and are merged into the single summary table that every analysis in this report is built on.

## SECTION 03

# Exploratory Data Analysis

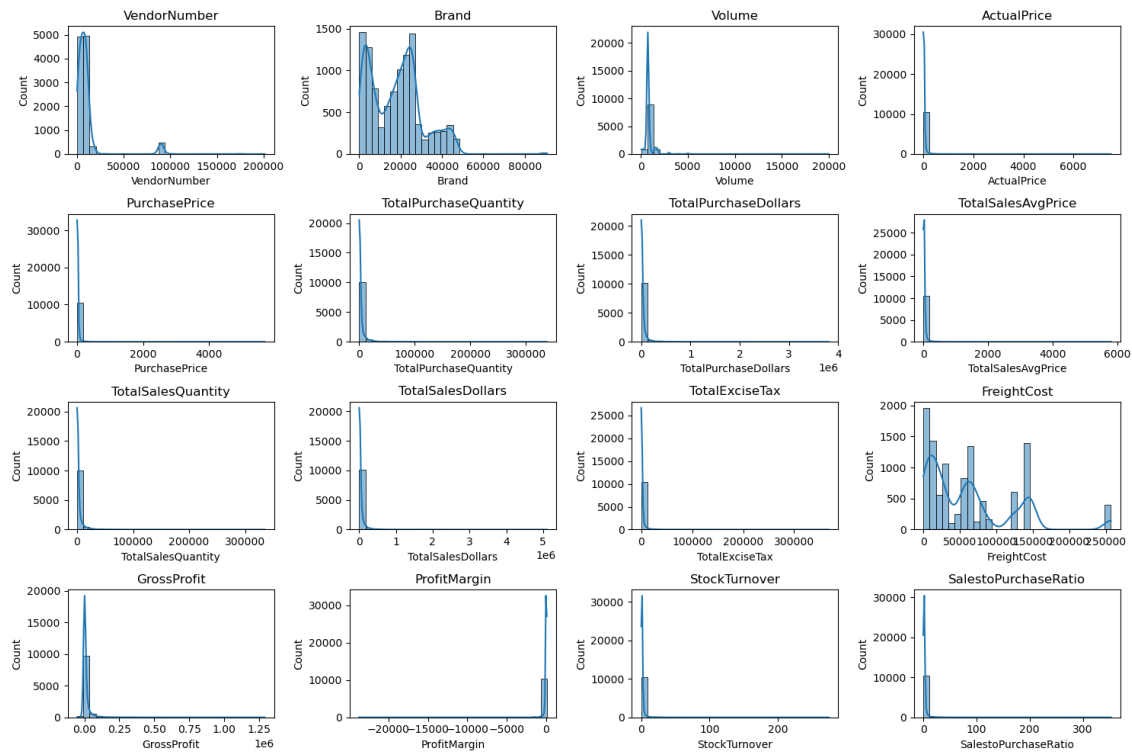
*Distributions, data quality checks, and the relationships between key metrics*

## 3.1 Statistical Summary

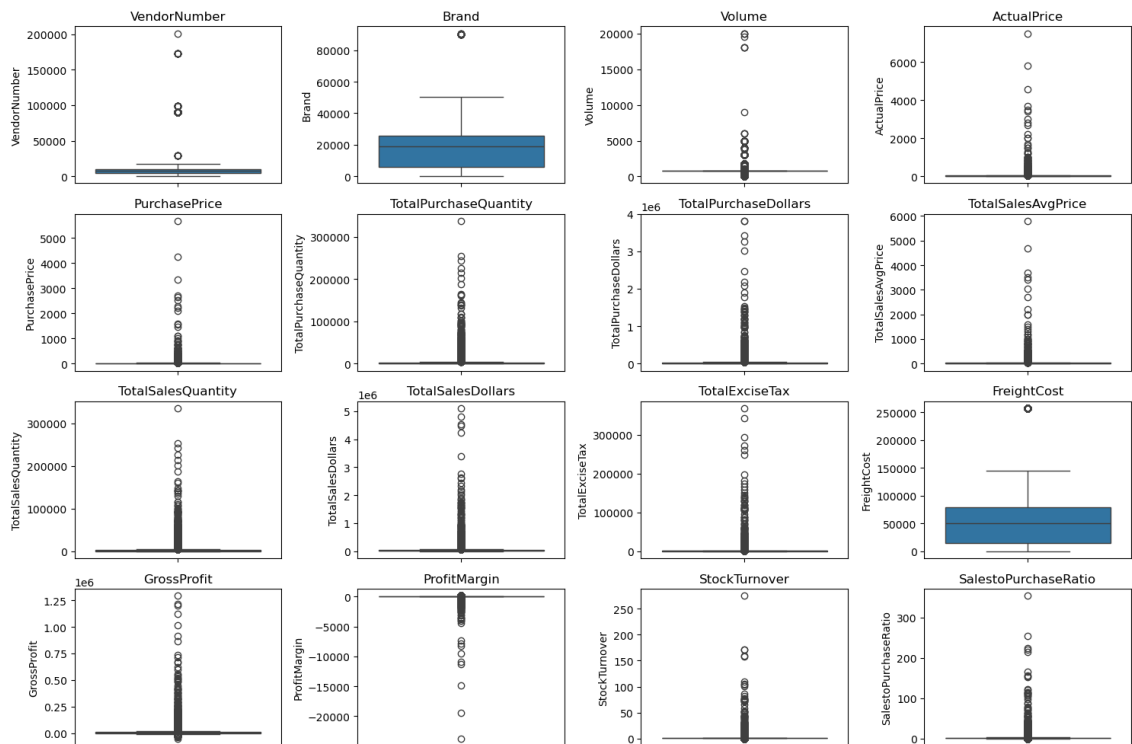
A first pass over the combined dataset surfaces its overall shape, scale, and the data-quality issues that need to be handled before any vendor-level conclusions can be trusted.

|                       | count    | mean     | std       | min       | 25%      | 50%      | 75%      | max        |
|-----------------------|----------|----------|-----------|-----------|----------|----------|----------|------------|
| VendorNumber          | 10692.00 | 10650.65 | 18753.52  | 2.00      | 3951.00  | 7153.00  | 9552.00  | 201359.00  |
| Brand                 | 10692.00 | 18039.23 | 12662.19  | 58.00     | 5793.50  | 18761.50 | 25514.25 | 90631.00   |
| Volume                | 10692.00 | 847.36   | 664.31    | 50.00     | 750.00   | 750.00   | 750.00   | 20000.00   |
| ActualPrice           | 10692.00 | 35.64    | 148.25    | 0.49      | 10.99    | 15.99    | 28.99    | 7499.99    |
| PurchasePrice         | 10692.00 | 24.39    | 109.27    | 0.36      | 6.84     | 10.46    | 19.48    | 5681.81    |
| TotalPurchaseQuantity | 10692.00 | 3140.89  | 11095.09  | 1.00      | 36.00    | 262.00   | 1975.75  | 337660.00  |
| TotalPurchaseDollars  | 10692.00 | 30106.69 | 123067.80 | 0.71      | 453.46   | 3655.47  | 20738.25 | 3811251.60 |
| TotalSalesAvgPrice    | 10692.00 | 33.23    | 125.26    | 0.00      | 10.16    | 15.15    | 27.66    | 5799.99    |
| TotalSalesQuantity    | 10692.00 | 3077.48  | 10952.85  | 0.00      | 33.00    | 261.00   | 1929.25  | 334939.00  |
| TotalSalesDollars     | 10692.00 | 42239.07 | 167655.27 | 0.00      | 729.22   | 5298.05  | 28396.92 | 5101919.51 |
| TotalExciseTax        | 10692.00 | 1774.23  | 10975.58  | 0.00      | 4.80     | 46.57    | 418.65   | 368242.80  |
| FreightCost           | 10692.00 | 61433.76 | 60938.46  | 0.09      | 14069.87 | 50293.62 | 79528.99 | 257032.07  |
| GrossProfit           | 10692.00 | 12132.38 | 46224.34  | -52002.78 | 52.92    | 1399.64  | 8660.20  | 1290667.91 |
| ProfitMargin          | 10692.00 | -inf     | NaN       | -inf      | 13.32    | 30.41    | 39.96    | 99.72      |
| StockTurnover         | 10692.00 | 1.71     | 6.02      | 0.00      | 0.81     | 0.98     | 1.04     | 274.50     |
| SalestoPurchaseRatio  | 10692.00 | 2.50     | 8.46      | 0.00      | 1.15     | 1.44     | 1.67     | 352.93     |

**FIGURE 1** Descriptive statistics (count, mean, std, min, quartiles, max) for every numeric field in vendor\_sales\_summary.



**FIGURE 2** Distribution of each numeric variable — most purchase and sales metrics are heavily right-skewed.



**FIGURE 3** Box plots for the same variables, exposing the extreme high-end outliers behind that skew.

### Negative & Zero Values

- Gross Profit: Minimum values are **-52,002.78**, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- Profit Margin: Has a minimum **-inf**, which suggests cases where revenue is zero or even lower than costs.
- Total Sales Quantity & Sales Dollars: Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or absolute stock.

### Outliers Indicated by High Standard Deviation

- Purchase Price & Actual Price: The max values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating potential premium products.
- Freight Cost: Huge variation, from 0.09 to 257,032.07, suggests logistics insufficiencies or bulk shipments.
- Stock Turnover: Ranges from 0 to 274.5, implying some products sell really fast while others remain in stock indefinitely. Value more than 1 indicates that sold quantity for that product is higher than purchased quantity due to either sales are being fulfilled from older stock.

## 3.2 Data Filtering

To improve the reliability of the insights, the following data points were removed:

### Exclusion Criteria

- Excluded records with zero sales because, without time-series data, we cannot determine whether the product is newly introduced or has been available for a while but is not selling.
- Excluded records where total sales quantity is higher than total purchase quantity because some of the sales may have come from older stock, which could make newly added products look less or more successful than they actually are.
- Excluded records with a profit margin of **-101% or lower** because such extreme losses may indicate (may vary business to business):
  - A newly introduced product which has just started selling, or
  - A large amount of unsold inventory, which businesses may simply clear out, discard, or return to the vendor rather than analyse further.

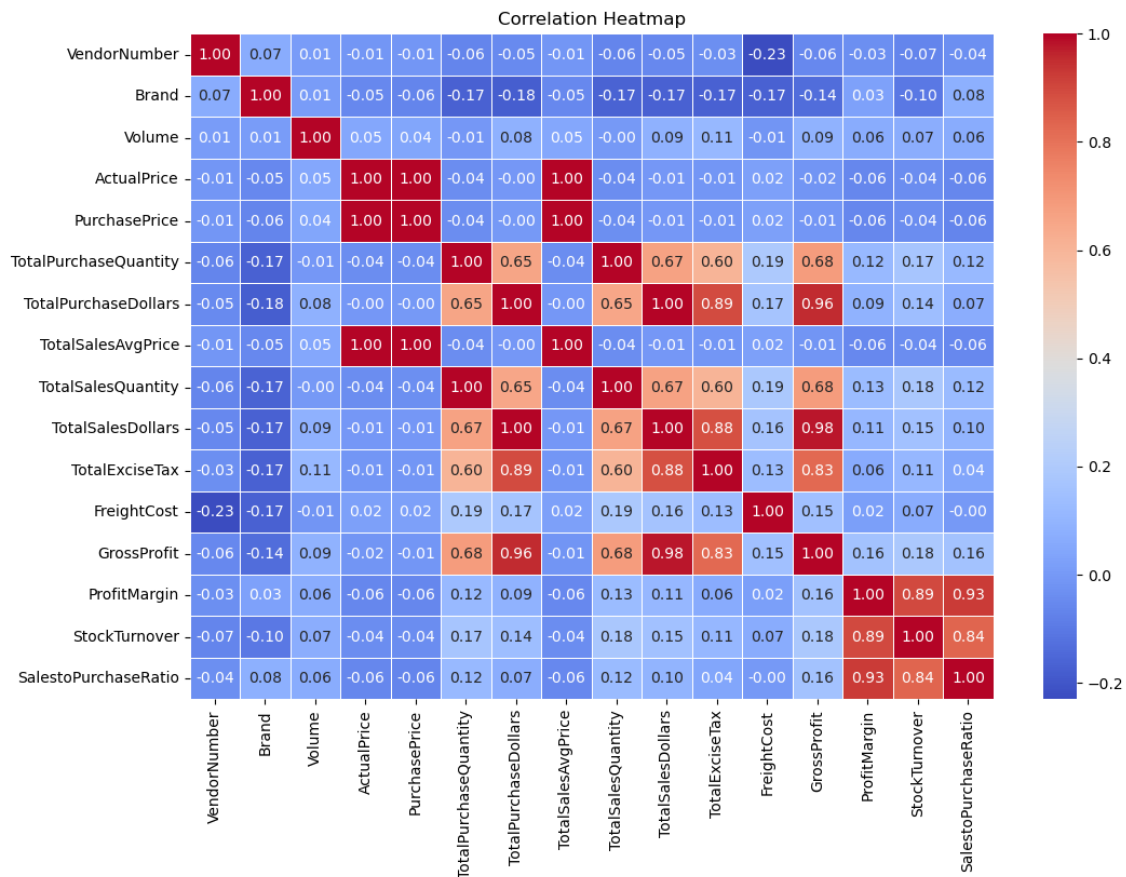
Keeping such records could therefore lead to misleading conclusions about product and vendor performance.

```
df_2 = pd.read_sql("""
SELECT *
FROM vendor_sales_summary
WHERE TotalSalesQuantity > 0
AND NOT (TotalSalesQuantity > TotalPurchaseQuantity)
AND ProfitMargin > -101
""", conn)
```

**FIGURE 4** SQL filter excluding zero-sales, stock-inflated, and extreme-loss records.



### 3.3 Correlation Insights



**FIGURE 5** Correlation heatmap across all numeric vendor, purchase, and sales metrics.

- PurchasePrice has weak negative correlation with TotalSalesDollars (-0.01) and GrossProfit (-0.01), suggesting that price variations do not significantly impact sales revenue or profit.
- Strong correlation between total purchase quantity and Total Sales Quantity (**1.00**), confirming efficient inventory turnover.
- Weak Negative correlation between profit margin & total sales average price (-0.06) suggests that, for some products as sales price increases, margins decrease, possibly due to competitive pricing pressures.
- Stock Turnover has positive correlations with both GrossProfit (0.18) and ProfitMargin (0.89), indicating that faster turnover necessarily results in higher profitability.
- Total Sales Quantity shows a strong positive correlation with Gross Profit (0.68), indicating that higher sales volumes are generally associated with higher gross profits.

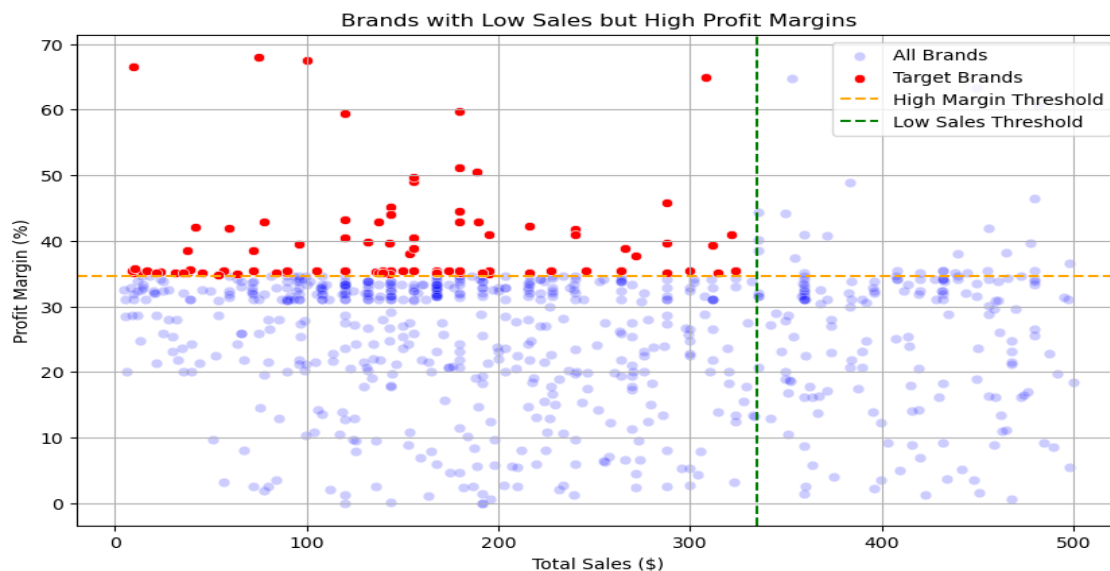
## 3.4 Research Questions & Key Findings

### 3.4.1 Brands with Lower Sales but Higher Profit Margins

|      | Description                  | TotalSalesDollars | ProfitMargin |
|------|------------------------------|-------------------|--------------|
| 4443 | Redtree Moscato              | 8.99              | 35.48        |
| 1198 | Ch Lestiac Cotes de Bordeaux | 9.49              | 35.09        |
| 4662 | Santa Rita Organic Svgn Bl   | 9.99              | 66.47        |
| 1208 | Ch Maycott Red               | 10.49             | 35.08        |
| 5760 | ZIPZ Pnt Grigio              | 10.74             | 35.75        |
| ...  | ...                          | ...               | ...          |
| 5543 | Viberti Chard                | 308.23            | 64.84        |
| 3045 | Kirkland Signature Cab Svgn  | 311.76            | 39.26        |
| 3018 | Kesselstatt Piesp G Kab Rsl  | 314.79            | 35.09        |
| 29   | 2 Gingers Irish Whiskey      | 321.86            | 40.89        |
| 3211 | Laurent Reverdy Sancerre     | 323.76            | 35.51        |

91 rows × 3 columns

**FIGURE 6** The identified low-sales, high-margin brands (91 rows), ranked by sales value.



**FIGURE 7** Sales value vs. profit margin — the low-sales, high-margin brands highlighted in red.

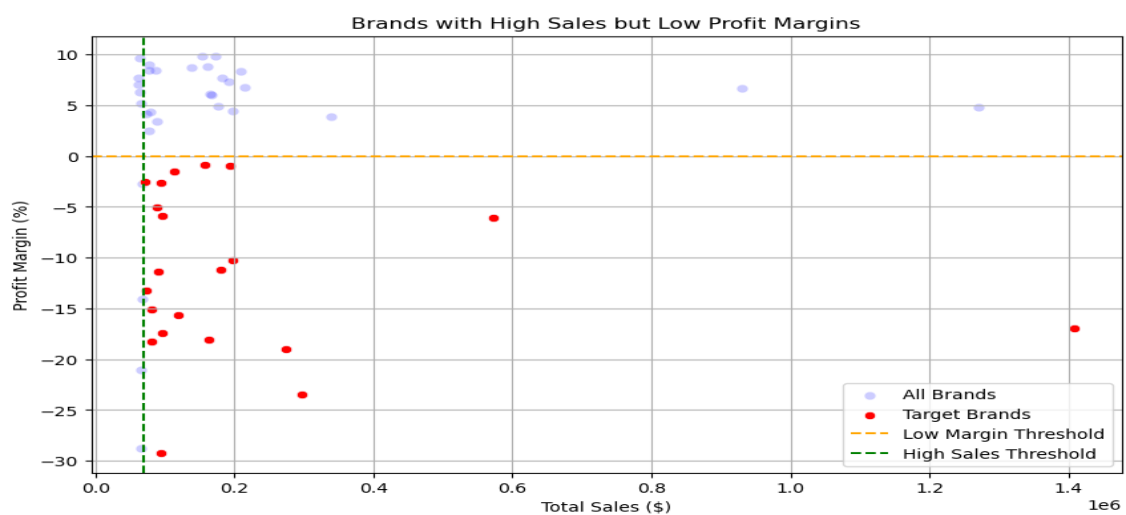
There are **91 brands** which have low sales but high profit margin. Targeted marketing, promotional campaigns or price adjustments may improve their sales volume without affecting their high profitability.

### 3.4.2 Brands with Higher Sales but Lower Profit Margins

|      | Description                  | TotalSalesDollars | ProfitMargin |
|------|------------------------------|-------------------|--------------|
| 4523 | Rocky Peak Hard Cinnamon     | 72075.00          | -2.56        |
| 4449 | Remy Martin 1738 Accord Royl | 72486.63          | -13.23       |
| 5130 | Tanteo Jalapeno Tequila      | 80547.53          | -18.22       |
| 4028 | OuterSpace Vodka             | 81795.02          | -15.12       |
| 2780 | Jack Daniels Barrel Proof    | 89296.41          | -5.08        |
| 2063 | Fernet-Branca Italian Liquer | 90170.16          | -11.39       |
| 4965 | St Remy VSOP Authentic Brndy | 94103.99          | -29.21       |
| 2028 | Fabrizia's Margarita RTD     | 94406.84          | -2.66        |
| 1385 | Clase Azul Reposado Tequila  | 95499.46          | -17.39       |
| 395  | Banfi Rosa Regale            | 96923.38          | -5.89        |
| 1556 | Crown Royal Nrth Harvest Rye | 113966.29         | -1.53        |
| 789  | Buehler Znfld Napa           | 119616.75         | -15.68       |
| 5515 | Vendange Chard               | 157952.39         | -0.88        |
| 3061 | Knob Creek Rye               | 162245.21         | -18.12       |
| 2374 | Glen Ellen Cab Svgn RSV      | 179932.54         | -11.18       |
| 1370 | Ciroc Coconut Vodka          | 192947.38         | -0.95        |
| 4214 | Pinnacle Tropical Punch      | 197259.99         | -10.22       |
| 5030 | Stolichnaya Razberi          | 274277.56         | -18.98       |
| 2873 | Jim Beam Black               | 296227.06         | -23.52       |
| 994  | Capt Morgan Spiced Rum 100   | 573230.06         | -6.08        |
| 5304 | Three Olives Vodka           | 1408861.65        | -16.97       |

21 rows, 3 columns

**FIGURE 8** The identified high-sales, low-margin brands (21 rows), ranked by sales value.



**FIGURE 9** Sales value vs. profit margin — the high-sales, low-margin brands highlighted in red.

There are **21 brands** which have high sales but low or negative profit margins. Since these brands have good sales, we should not discontinue them just because they are not profitable. Instead, we should discuss with vendors and look for ways to reduce their purchase prices. This can help improve the profit margins while maintaining the good sales of these brands.

### 3.4.3 Vendors & Brands with the Highest Sales Performance

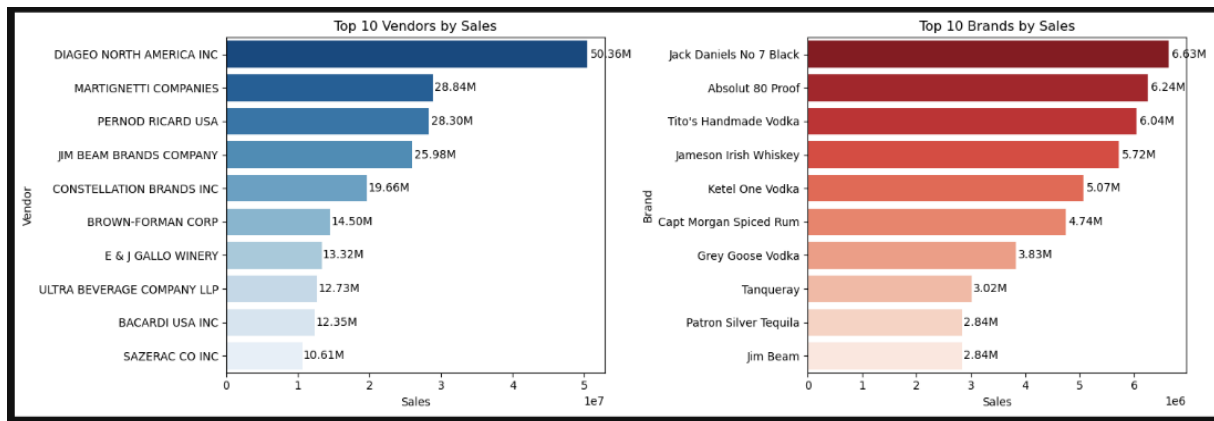


FIGURE 10 Top vendors and brands ranked by total sales value.

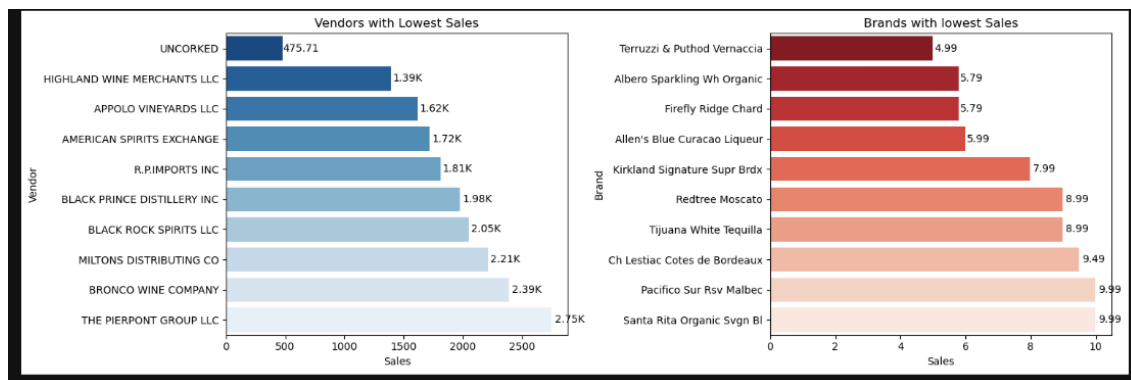
#### Top Vendors

- The business should maintain strong and long-term relationships with these top vendors, as they contribute a major share of the business's total sales. Any disruption or loss of even one of these important vendors could significantly reduce sales and affect overall business performance.

#### Top Brands

- The business should ensure that these top-selling brands always have sufficient stock to meet customer demand and avoid losing potential sales. At the same time, low-performing brands should be promoted strategically to improve their sales without affecting the availability and sales of these high-performing brands.

### 3.4.4 Vendors & Brands with the Lowest Sales Performance



**FIGURE 11** Bottom vendors and brands ranked by total sales value.

#### Low-Sales Vendors

- The business should review these vendors and reduce purchases from them if their products have consistently low demand. This can help avoid investing money in slow-moving products.

#### Low-Sales Brands

- These brands should be reviewed based on their sales and profitability. If they continue to perform poorly, their stock can be reduced or they can be discontinued, allowing more focus on better-performing brands.

### 3.4.5 Purchase Concentration Among Top Vendors

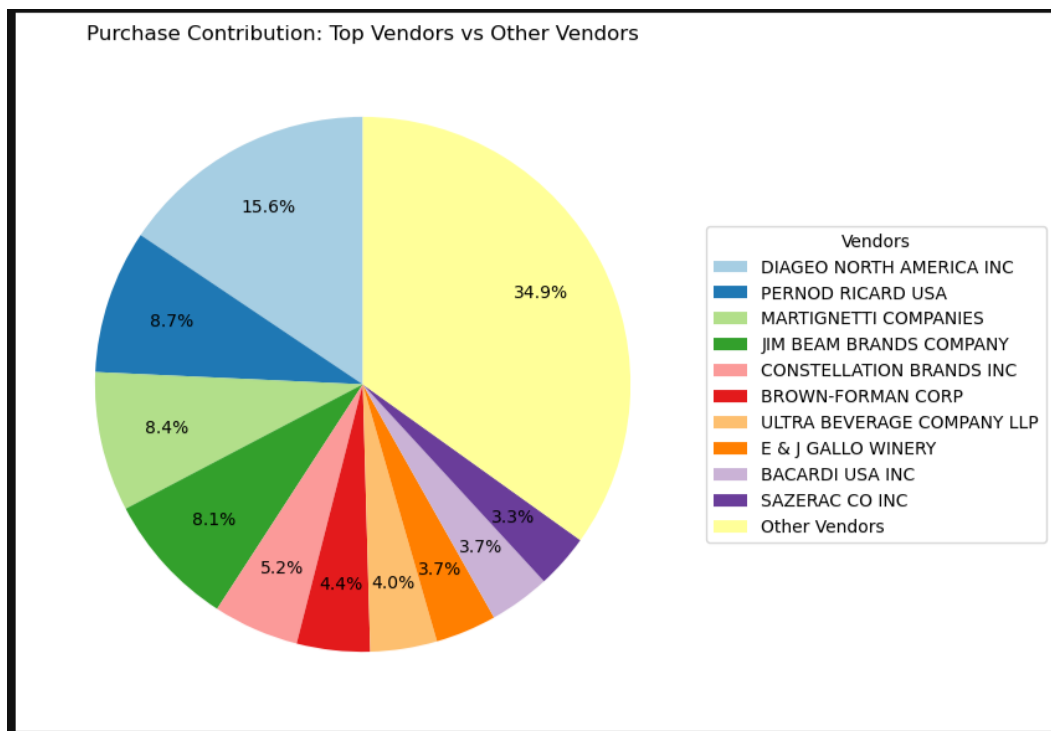


FIGURE 12 Purchase contribution — top 10 vendors vs. all other vendors.

|     | VendorName                 | TotalPurchaseDollars | GrossProfit | TotalSalesDollars | PurchaseContribution% | Cumulative_Contribution |
|-----|----------------------------|----------------------|-------------|-------------------|-----------------------|-------------------------|
| 25  | DIAGEO NORTH AMERICA INC   | 38.31M               | 12.06M      | 50.36M            | 15.57                 | 15.57                   |
| 66  | PERNOD RICARD USA          | 21.38M               | 6.92M       | 28.30M            | 8.69                  | 24.27                   |
| 55  | MARTIGNETTI COMPANIES      | 20.73M               | 8.10M       | 28.84M            | 8.43                  | 32.70                   |
| 44  | JIM BEAM BRANDS COMPANY    | 20.04M               | 5.94M       | 25.98M            | 8.15                  | 40.84                   |
| 20  | CONSTELLATION BRANDS INC   | 12.75M               | 6.92M       | 19.66M            | 5.18                  | 46.03                   |
| 12  | BROWN-FORMAN CORP          | 10.87M               | 3.63M       | 14.50M            | 4.42                  | 50.44                   |
| 103 | ULTRA BEVERAGE COMPANY LLP | 9.91M                | 2.82M       | 12.73M            | 4.03                  | 54.48                   |
| 30  | E & J GALLO WINERY         | 9.06M                | 4.26M       | 13.32M            | 3.68                  | 58.16                   |
| 7   | BACARDI USA INC            | 9.03M                | 3.31M       | 12.35M            | 3.67                  | 61.83                   |
| 77  | SAZERAC CO INC             | 8.05M                | 2.56M       | 10.61M            | 3.27                  | 65.10                   |

FIGURE 13 Purchase share by vendor, ranked.

These top 10 vendors contribute around **65.10%** of total purchases, making them a major source of our inventory. The business should maintain strong relationships with these vendors, as losing even one of them could create a significant shortage in inventory. At the same time, the business should gradually increase purchases from other reliable vendors to reduce its dependence on these top 10 vendors and lower the supply risk.

### 3.4.6 Impact of Bulk Purchasing on Unit Price

```
OrderSize
Small    32.01
Medium   17.37
Large    11.18
Name: UnitPurchasePrice, dtype: float64
```

FIGURE 14 Average unit purchase price by order size.

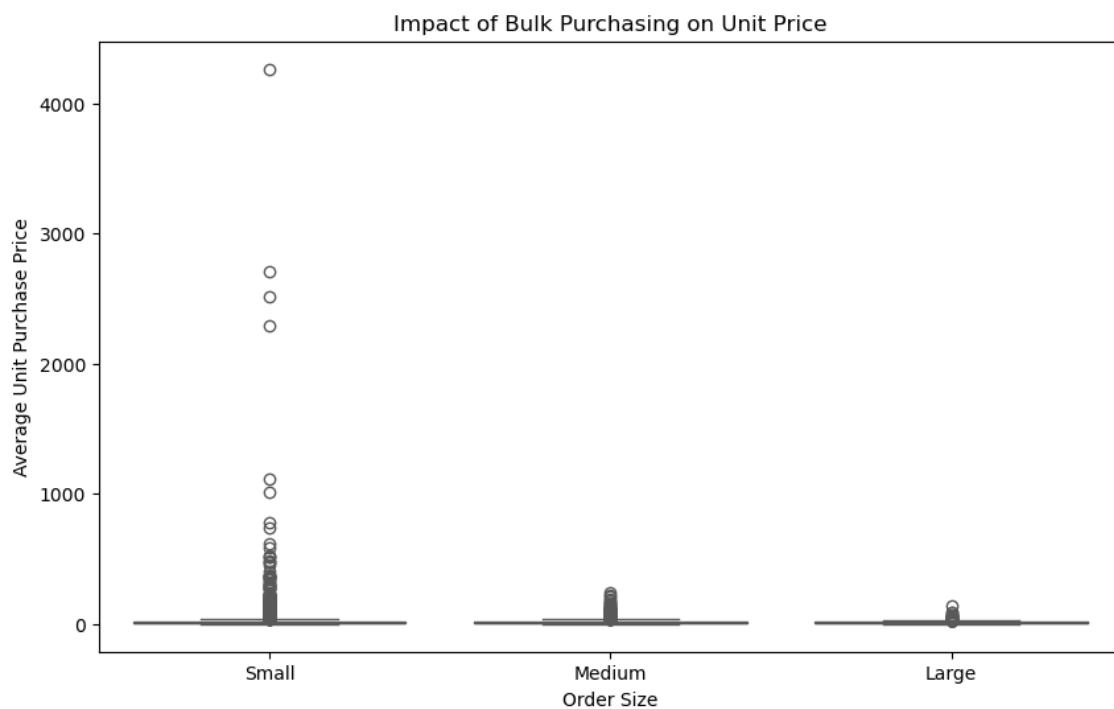


FIGURE 15 Unit purchase price across order-size bands.

- Buying products in bulk (*Large Order Size*) from vendors leads to lowest average unit price (**\$11.18 per unit**), meaning higher margins if business can manage inventory efficiently.
- Larger order sizes result in lower per-unit purchase costs, indicating that bulk purchasing allows the business to achieve better pricing from vendors.

### 3.4.7 Vendors with the Lowest Inventory Turnover

Top 10 vendors with the lowest Stock Turnover :

|   | VendorName                  | StockTurnover |
|---|-----------------------------|---------------|
| 0 | UNCORKED                    | 0.36          |
| 1 | VINEYARD BRANDS LLC         | 0.44          |
| 2 | Russian Standard Vodka      | 0.45          |
| 3 | GILMANTON WINERY & VINEYARD | 0.47          |
| 4 | APPOLO VINEYARDS LLC        | 0.47          |
| 5 | Serralles Usa LLC           | 0.53          |
| 6 | SOUTHERN GLAZERS W&S OF NE  | 0.60          |
| 7 | NICHE W & S                 | 0.61          |
| 8 | HIGHLAND WINE MERCHANTS LLC | 0.61          |
| 9 | ALISA CARR BEVERAGES        | 0.63          |

**FIGURE 16** Vendors ranked by lowest inventory turnover.



### 3.4.8 Vendors with the Highest Unsold Inventory Value

**Total Unsold Capital : 13.21M**

**FIGURE 17** Total unsold capital tied up across vendors.

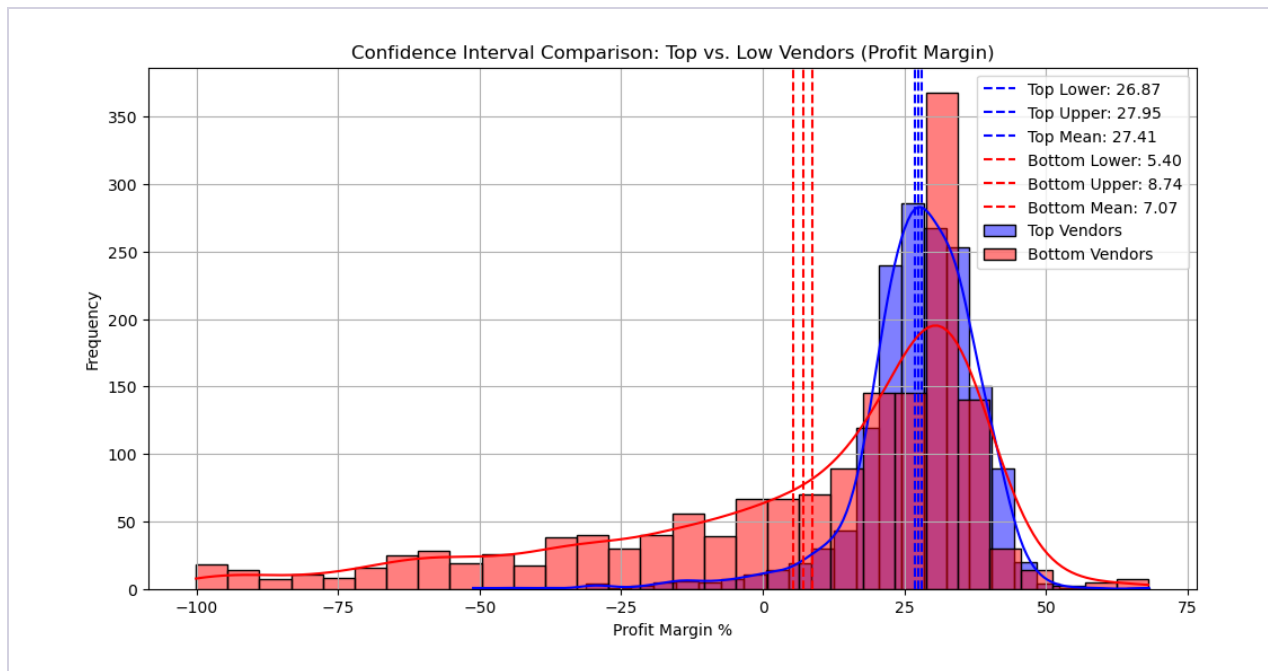
Top 10 vendors with the highest Unsold Inventory Value:

|     | VendorName                 | UnsoldInventoryValue | UnsoldInventoryContribution% |
|-----|----------------------------|----------------------|------------------------------|
| 25  | DIAGEO NORTH AMERICA INC   | 1.60M                | 12.09                        |
| 55  | MARTIGNETTI COMPANIES      | 1.53M                | 11.58                        |
| 44  | JIM BEAM BRANDS COMPANY    | 1.02M                | 7.74                         |
| 103 | ULTRA BEVERAGE COMPANY LLP | 1.00M                | 7.58                         |
| 66  | PERNOD RICARD USA          | 708.17K              | 5.36                         |
| 65  | PERFECTA WINES             | 627.36K              | 4.75                         |
| 51  | M S WALKER INC             | 529.60K              | 4.01                         |
| 113 | WILLIAM GRANT & SONS INC   | 492.70K              | 3.73                         |
| 30  | E & J GALLO WINERY         | 458.33K              | 3.47                         |
| 20  | CONSTELLATION BRANDS INC   | 431.59K              | 3.27                         |

The vendor DIAGEO NORTH AMERICA INC holds the highest Unsold Inventory Value of \$1.60M, contributing 12.09% to the total Unsold Inventory Value of \$13.21M.

**FIGURE 18** Vendors holding the highest unsold inventory value.

### 3.4.9 Profit Margin Gap: Low- vs. High-Performing Vendors



**FIGURE 19** Confidence intervals for profit margin — low-performing vs. top-performing vendors.

#### Interpretation

- The confidence interval for low-performing vendors (**5.40% to 8.74%**) is significantly lower than that of top-performing vendors (**26.87% to 27.95%**). This suggests that sales from low performing vendors may generate low profit margins due to their lower sales volumes, indicating that sales volume leads to higher profitability.
- For High-Performing vendors: We can try to implement some discounts or marketing campaigns to uplift the sales for their products more, which could result in higher Gross Profit as correlation between TotalSalesDollars and GrossProfit is 0.98.
- For Low-Performing vendors: These vendors have low profit margins, because of their lower sales. This could be because of customer interests, quality issues or due to some other factors, the responsible team should investigate into this.

### 3.4.10 Statistical Validation of the Profit Margin Difference

#### Hypothesis Test

Null Hypothesis: There is no significance difference in the mean profit margins of top-performing and low-performing vendors.

Alternate Hypothesis: The mean profit margins of top-performing and low-performing vendors are significantly different.

Result: **There is a significant difference in profit margins between top performing vendors and low performing vendors, Reject Null Hypothesis.**

Implication: High-performing vendors may achieve better profit margins due to the larger quantity of products they sell compared to Low-Performing vendors.

## SECTION 04

# Power BI Dashboard Overview

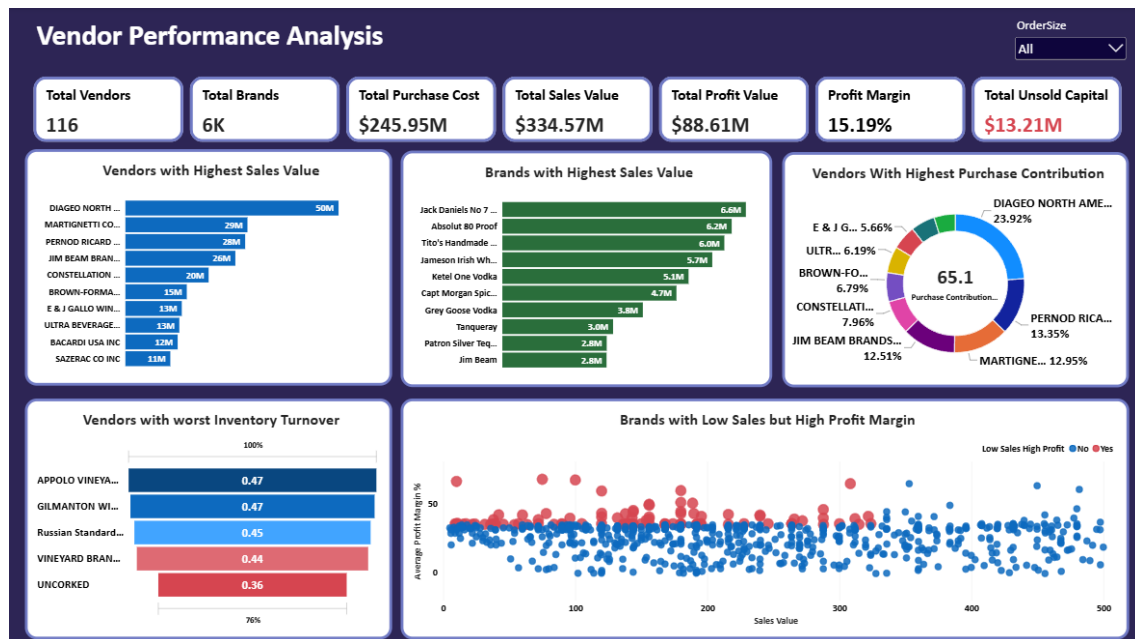
*All insights above, consolidated into two interactive dashboard pages*

Every analysis, insight, and visual above has been consolidated into an interactive Power BI dashboard for stakeholders — giving a clear, at-a-glance view of sales, vendor performance, profitability, purchasing, and inventory to support data-driven decisions.

## 4.1 Primary KPIs

|                                       |                                      |                                         |
|---------------------------------------|--------------------------------------|-----------------------------------------|
| <b>116</b><br>TOTAL VENDORS           | <b>6K</b><br>TOTAL BRANDS            | <b>\$245.95M</b><br>TOTAL PURCHASE COST |
| <b>\$334.57M</b><br>TOTAL SALES VALUE | <b>15.19%</b><br>TOTAL PROFIT MARGIN | <b>\$13.21M</b><br>TOTAL UNSOLD CAPITAL |

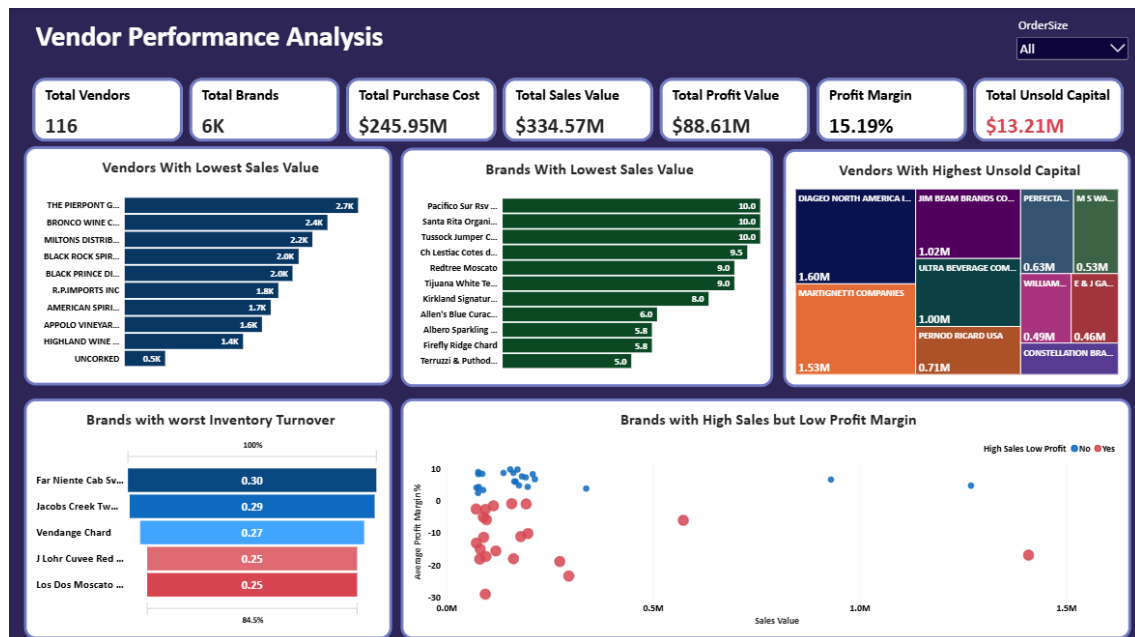
## 4.2 Dashboard — Page 1



**FIGURE 20** Dashboard page 1 — sales, purchase contribution, and inventory-turnover overview.

- Vendors With Highest Sales Value — top 10 vendors by total sales (*bar chart*)
- Brands With Highest Sales Value — top 10 brands by total sales (*bar chart*)
- Vendors With Highest Purchase Contribution — share of the top 10 purchasing vendors (*donut chart*)
- Vendors With Worst Inventory Turnover — 5 vendors with the lowest turnover (*funnel chart*)
- Brands With Low Sales but High Profit Margin — the 91-brand opportunity set (*scatter chart*)

## 4.3 Dashboard — Page 2



**FIGURE 21** Dashboard page 2 — lowest performers, unsold capital, and margin trade-offs.

- Vendors With Lowest Sales Value — bottom 10 vendors by total sales (*bar chart*)
- Brands With Lowest Sales Value — bottom 10 brands by total sales (*bar chart*)
- Vendors With Highest Unsold Capital — 10 vendors holding the most unsold value (*treemap*)
- Brands With Worst Inventory Turnover — 5 brands with the lowest turnover (*funnel chart*)
- Brands With High Sales but Low Profit Margin — the 21-brand renegotiation set (*scatter chart*)

## SECTION 05

# Final Recommendations

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*Turning the findings above into concrete next steps*

- 01** Maintain strong relationships with top vendors, as they supply the majority of purchases and inventory.
- 02** Negotiate lower purchase prices for high-selling, low-profit brands to improve their margins.
- 03** Keep sufficient stock of top-selling brands so that we don't lose sales due to shortages.
- 04** Reduce purchases of slow-moving products to limit capital tied up in unsold inventory.
- 05** Use bulk purchasing where possible to secure lower unit prices from vendors.
- 06** Reduce unsold inventory through discounts, promotions, or vendor returns.
- 07** Promote low-selling, high-profit brands to grow sales without diluting profitability.
- 08** Increase purchases from other reliable vendors to reduce dependence on the top 10.
- 09** Improve inventory turnover by focusing on products that sell slowly and finding ways to increase their sales.